

Topic 1: Basic Linux Commands

```
student@DESKTOP-UIMU3UD:~$ pwd
/home/student
```

```
student@DESKTOP-UIMU3UD:~$
```

```
student@DESKTOP-UIMU3UD:~$ history
 1 nano Hello.c
 2 pwd
 3 clear
 4 history
 5 whoami
 6 date
 7 cal
 8 sudo apt install ncal
 9 pwd
10 clear
11 pwd
12 clear
13 history
```

```
student@DESKTOP-UIMU3UD:~$ whoami
student
student@DESKTOP-UIMU3UD:~$ date
Sun Aug  9 08:27:32 UTC 2026
```

```
student@DESKTOP-UIMU3UD:~$ cal
      August 2026
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31
student@DESKTOP-UIMU3UD:~$ hostname
DESKTOP-UIMU3UD
```

```
student@DESKTOP-UIMU3UD:~$ uname -a
Linux DESKTOP-UIMU3UD 4.4.0-19041-Microsoft #4355-Microsoft Thu Apr 12 17:37:00 PST 2024 x86_64 x86_64 x86_64 GNU/Linux
```

Topic 2: Directory Management

```

student@DESKTOP-UIMU3UD:~$ ls
Hello.c      exec_example.c  fork_exec2      fork_test.c     name.c          parent_child_pid  pid.c          zombie
Hello.c.save fork_exec       fork_exec2.c    hello           parent_child     parent_child_pid.c  wait_example  zombie.c
exec_example  fork_exec.c     fork_test       name            parent_child.c  pid               wait_example.c
student@DESKTOP-UIMU3UD:~$ ls -l
total 176
-rw-r--r-- 1 student student 76 Aug 8 15:19 Hello.c
-rw-r--r-- 1 student student 155 Aug 8 15:14 Hello.c.save
-rwxr-xr-x 1 student student 16008 Aug 9 06:07 exec_example
-rw-r--r-- 1 student student 159 Aug 9 06:07 exec_example.c
-rwxr-xr-x 1 student student 16088 Aug 9 05:23 fork_exec
-rw-r--r-- 1 student student 259 Aug 9 05:23 fork_exec.c
-rwxr-xr-x 1 student student 16088 Aug 9 06:10 fork_exec2
-rw-r--r-- 1 student student 264 Aug 9 06:09 fork_exec2.c
-rwxr-xr-x 1 student student 16008 Aug 9 05:17 fork_test
-rw-r--r-- 1 student student 101 Aug 9 05:17 fork_test.c
-rwxr-xr-x 1 student student 15960 Aug 8 15:20 hello
-rwxr-xr-x 1 student student 16064 Aug 8 15:26 name
-rw-r--r-- 1 student student 154 Aug 8 15:25 name.c
-rwxr-xr-x 1 student student 16008 Aug 9 05:21 parent_child
-rw-r--r-- 1 student student 226 Aug 9 05:20 parent_child.c
-rwxr-xr-x 1 student student 16104 Aug 9 06:02 parent_child_pid
-rw-r--r-- 1 student student 287 Aug 9 06:01 parent_child_pid.c
-rwxr-xr-x 1 student student 16000 Aug 8 15:28 pid
-rw-r--r-- 1 student student 108 Aug 8 15:28 pid.c
-rwxr-xr-x 1 student student 16048 Aug 9 06:04 wait_example
-rw-r--r-- 1 student student 273 Aug 9 06:04 wait_example.c
-rwxr-xr-x 1 student student 16088 Aug 9 05:26 zombie
-rw-r--r-- 1 student student 303 Aug 9 05:26 zombie.c
student@DESKTOP-UIMU3UD:~$ ls -a
.      .local      exec_example  fork_test     parent_child.c  wait_example.c
..     .motd shown exec_example.c fork_test.c    parent_child_pid  zombie
.bash_history .profile    fork_exec     hello         parent_child_pid.c  zombie.c
.bash_logout .sudo_as_admin_successful fork_exec.c    name          pid
.bashrc      Hello.c     fork_exec2    name.c        pid.c
.landscape   Hello.c.save fork_exec2.c   parent_child  wait_example
student@DESKTOP-UIMU3UD:~$ mkdir OSLab
student@DESKTOP-UIMU3UD:~$ cd OSLab
student@DESKTOP-UIMU3UD:~/OSLab$ cd ..
student@DESKTOP-UIMU3UD:~$ rmdir Test
rmdir: failed to remove 'Test': No such file or directory

```

```

student@DESKTOP-UIMU3UD:~$ find . -name "*.c"
./exec_example.c
./fork_exec.c
./fork_exec2.c
./fork_test.c
./Hello.c
./name.c
./parent_child.c
./parent_child_pid.c
./pid.c
./wait_example.c
./zombie.c

```

```

student@DESKTOP-UIMU3UD:~$ locate hello.c
student@DESKTOP-UIMU3UD:~$ locate hello.c
student@DESKTOP-UIMU3UD:~$ locate hello.c
student@DESKTOP-UIMU3UD:~$

```

Topic 3: File Management

```

student@DESKTOP-UIMU3UD:~$ touch file1.txt
student@DESKTOP-UIMU3UD:~$ cat file1.txt
student@DESKTOP-UIMU3UD:~$ cp file1.txt file2.txt
student@DESKTOP-UIMU3UD:~$ mv file2.txt renamed.txt
student@DESKTOP-UIMU3UD:~$ rm renamed.txt
student@DESKTOP-UIMU3UD:~$ head Hello.c
#include<stdio.h>
int main()
{
    printf("Hello Linux\n");
    return 0;
}
student@DESKTOP-UIMU3UD:~$ tail Hello.c
#include<stdio.h>
int main()
{
    printf("Hello Linux\n");
    return 0;
}
student@DESKTOP-UIMU3UD:~$ more Hello.c
#include<stdio.h>
int main()
{
    printf("Hello Linux\n");
    return 0;
}

```

```

student@DESKTOP-UIMU3UD:~$ less Hello.c
student@DESKTOP-UIMU3UD:~$ wc Hello.c
  6  9 76 Hello.c
student@DESKTOP-UIMU3UD:~$ diff Hello.c name.c
4c4,7
<     printf("Hello Linux\n");
---
>     char name[30];
>     printf("Enter your name: ");
>     scanf("%s",name);
>     printf("Welcome %s\n",name);
student@DESKTOP-UIMU3UD:~$ sort file1.txt
student@DESKTOP-UIMU3UD:~$ uniq file1.txt
student@DESKTOP-UIMU3UD:~$ grep main Hello.c
int main()
student@DESKTOP-UIMU3UD:~$

```

Topic 4: File Editing

```

student@DESKTOP-UIMU3UD:~$ nano hello.c
student@DESKTOP-UIMU3UD:~$ vi test.c
student@DESKTOP-UIMU3UD:~$ nano hello.c
student@DESKTOP-UIMU3UD:~$ vi test.c
student@DESKTOP-UIMU3UD:~$ history | tail -n 5
 49  nano hello.c
 50  vi test.c
 51  nano hello.c
 52  vi test.c
 53  history | tail -n 5
student@DESKTOP-UIMU3UD:~$

```

Topic 5: C Program Compilation and Execution

```

=== Topic 5: C Program Compilation and Execution ===
Hello World
/usr/bin/gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc.1.gz

```

Topic 6: Process Management

```

=== Topic 6: Process Management ===
  PID TTY          TIME CMD
   83 tty1        00:00:00 bash
  101 tty1        00:00:00 run_all.sh
  140 tty1        00:00:00 ps
UID      PID  PPID  C STIME TTY          TIME CMD
root         1    0  0 11:35 ?        00:00:00 /init
root         5    1  0 11:35 ?        00:00:00 plan9 --control-socket 6 --log-level 4 --serv
er-fd 7 --pipe-fd 9 --socket-path /mnt/c/Users/Abinash/AppData/Local/Packages/CanonicalGroupL
imited.Ubuntu_79rhkp1fndgsc/LocalState/fsserver --log-truncate
root        82    1  0 11:38 tty1        00:00:00 /init
student     83    82  0 11:38 tty1        00:00:00 -bash
student    101    83  2 11:39 tty1        00:00:00 /bin/bash ./run_all.sh
student    141   101  0 11:39 tty1        00:00:00 ps -ef
student    142   101  0 11:39 tty1        00:00:00 head -n 10

```

Topic 7: Job Control

```

=== Topic 7: Job Control ===
[1]+  Running                  ./hello &

```

Topic 8: File Permissions and Links

```

=== Topic 8: File Permissions and Links ===
Hello World

```

Topic 9: Disk and Memory Information

```

=== Topic 9: Disk and Memory Information ===
Filesystem      Size  Used Avail Use% Mounted on
rootfs          248G   74G  175G  30% /
none            248G   74G  175G  30% /dev
none            248G   74G  175G  30% /run
none            248G   74G  175G  30% /run/lock
none            248G   74G  175G  30% /run/shm
none            248G   74G  175G  30% /run/user
tmpfs           248G   74G  175G  30% /sys/fs/cgroup
C:\             248G   74G  175G  30% /mnt/c
D:\            196G   43G  153G  22% /mnt/d
224K      .

```

	total	used	free	shared	buff/cache	available
Mem:	3.9Gi	3.0Gi	915Mi	17Mi	223Mi	915Mi
Swap:	12Gi	386Mi	11Gi			

Topic 10: User and Environment Management

```

=== Topic 10: User and Environment Management ===
uid=1000(student) gid=1000(student) groups=1000(student),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),107(netdev)SHELL=/bin/bash
WSL_DISTRO_NAME=Ubuntu
NAME=DESKTOP-UIMU3UD
PWD=/home/student
LOGNAME=student
HOME=/home/student
LANG=C.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arj=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.taz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:~=00;90:*=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpkg-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color

```